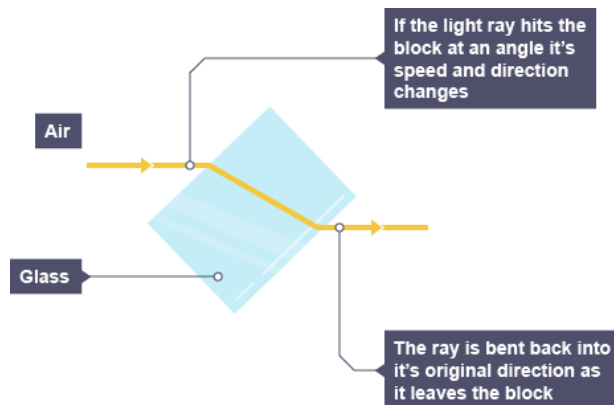


Practical Skills in Agriculture

Testing for the Sugar Content of Food Using the Physics of Light

Light is bent every time it comes through the lenses of our eyes; and every time it goes through one substance like water or air, and then goes into another substance like glass or plastic.



The amount of dissolved substances in a solution bends light by different amounts. Sugar is a specific example of this. We can test how much sugar food has in it by seeing how much light can be bent by testing the juice of fruits and vegetables.

Farmers can work out when their fruit is ripe - so they can pick it at just the right time to eat; or make wine from the fruit or preserve it in some way.

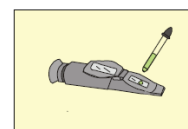
A very common practical skill is to do this testing with a device called a refractometer. Below is a picture of one. The refractometer measures the amount of sugar in what are called **Brix units**. Higher values mean higher nutrition value.

The procedure for doing the testing is shown on the right.

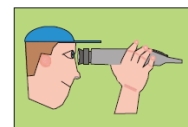


Above is the refractometer similar to the one we will use in class.

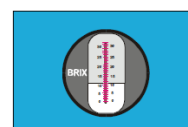
**BRIX TESTING IS INCREDIBLY
SIMPLE AND THE RESULT IS
INSTANTANEOUS!**



1. Squeeze a drop



2. Peek at the screen



3. Check the chart

Experiment

Today you will measure the sugar content of dissolved substances in different fruits and vegetables. First, you need to calibrate your refractometer by using distilled water. You may have to zero the device by adjusting the top screw, but seek your teacher's advice if you need to do this.

Next, the fruits and vegetables must be juiced by using either a juice extractor or a simple fork and squashing the fruit. Then a sample of the juice is placed on the refractometer, the flap closed and the measurement taken.

Between each new sample, the delicate refractometer plate must be wiped clean with distilled water (that has no dissolved substances in it).

Record your results below – Get each member of your group to take the reading

Food Item	Brix Reading 1	Brix Reading 2	Brix Reading 3	Average
Dissolved sugar				

1. Which of your food has the highest brix measurement? _____

2. Compare your results for at least two food items with typical measurements taken from them

Food 1: _____ Your Reading _____ Poor, Average, Good, Excellent (circle)

Food 2: _____ Your Reading _____ Poor, Average, Good, Excellent (circle)

**Refractive Index of Crop Juices Calibrated
in % Sucrose or Degree Brix**

VEGETABLES				
Vegetables	Poor	Average	Good	Excellent
Asparagus	2	4	6	8
Beets	6	8	10	12
Bell Peppers	4	6	8	12
Broccoli	6	8	10	12
Cabbage	6	8	10	12
Carrots	4	6	12	18
Cauliflower	4	6	8	10
Celery	4	6	10	12
Corn Stalks	4	8	14	20
Corn (Young)	6	10	18	24
Cow Peas	4	6	10	12
Endive	4	6	8	10
English Peas	8	10	12	14
Escarole	4	6	8	10
Field Peas	4	6	10	12
Green Beans	4	6	8	10
Hot Peppers	4	6	8	10
Kohlrabi	6	8	10	12
Lettuce	4	6	8	10
Onions	4	6	8	10
Parsley	4	6	8	10
Peanuts	4	6	8	10
Potatoes, Irish	3	5	7	8
Potatoes, Red	3	5	7	8
Potatoes, Sweet	6	8	10	14
Romaine	4	6	8	10
Rutabagas	4	6	10	12
Squash	6	8	12	14
Sweet Corn	6	10	18	24
Turnips	4	6	8	10

**Refractive Index of Crop Juices Calibrated
in % Sucrose or Degree Brix**

FRUITS - GRASSES				
Fruits	Poor	Average	Good	Excellent
Apple	6	10	14	18
Avocadoes	4	6	8	10
Bananas	8	10	12	14
Cantaloupe	8	12	14	16
Casaba	8	10	12	14
Cherries	6	8	14	16
Coconut	8	10	12	14
Grapes	8	12	14	20
Grapefruit	6	10	14	18
Honeydew	8	10	12	14
Kumquat	4	6	8	10
Lemons	4	6	8	12
Limes	4	6	10	12
Mangos	4	6	10	14
Oranges	6	10	16	20
Papayas	6	10	18	22
Peaches	6	10	14	18
Pears	6	10	12	14
Pineapple	12	14	20	22
Raisins	60	70	75	80
Raspberry	6	8	12	14
Strawberry	6	10	14	16
Tomato	4	6	8	12
Watermelon	4	6	8	12
Grasses				
Alfalfa	4	8	16	22
Grains	6	10	14	18
Sorghum	6	10	22	30